Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

Michael Fang
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Primary advisor: Professor Salvatore Torquato
Second reader: TBD

This paper represents my own work in accordance with University regulations.
/s/ Michael Fang

Abstract

We present a systematic numerical study of hyperuniformity in one-dimensional point patterns, characterizing 27 systems that span all three universality classes. Number variance, structure factor, and diffusion spreadability serve as complementary probes of long-wavelength density fluctuations, validated against exact analytical benchmarks before application to the full pattern catalog. We show that all five metallic-mean quasicrystal chains (Fibonacci through Nickel) are constrained to the hyperuniformity exponent $\alpha = 3$ by the unimodular determinant condition on their 2×2 substitution matrices, and report surface-area coefficients $\bar{\Lambda}$ for all five chains, four of which are novel. Moving to 3×3 matrices, we identify the Bombieri-Taylor cubic substitution as the first known one-dimensional quasicrystal with $1 < \alpha < 2$, obtaining $\alpha = 1.545$ in quantitative agreement with the eigenvalue formula of Oğuz et al. An exhaustive enumeration of 39,304 candidate 3×3 non-negative integer matrices proves algebraically that the interval $2 < \alpha < 3$ is inaccessible at this matrix rank, while yielding a new deterministic substitution tiling with $\alpha = 2$ exactly — to our knowledge the first quasiperiodic structural analogue of the stochastic uncorrelated random lattice. Generalized scaling coefficients are introduced to rank Class II and III patterns, and stealthy hyperuniform configurations are found to exist in two distinct structural phases that both satisfy the stealthy constraint while differing substantially in short-range order.

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1 Introduction

Long-range density fluctuations are a central diagnostic of order in many-body systems. When points are distributed at random with no correlations, the variance of the count inside a window of half-width R grows in proportion to the window length: $\sigma^2_{\text{Poisson}}(R) = 2\rho R$, where ρ is the mean density. This linear growth is the Poisson baseline — the generic behavior of uncorrelated disordered matter. *Hyperuniform* systems are those for which large-scale density fluctuations are anomalously suppressed relative to this baseline [1, 2]. Hyperuniformity is not a statement about short-range structure: a hyperuniform pattern can appear locally disordered while being globally organized in a precise statistical sense. This combination of apparent local randomness with long-range suppression of density fluctuations places hyperuniform systems in a structurally distinct class, intermediate between perfect crystals and uncorrelated fluids.

Hyperuniform arrangements appear across a remarkable range of physical and biological contexts. The photoreceptor mosaic of bird retinas forms a hyperuniform pattern, a design that enables near-uniform visual sampling without the periodic artifacts of a crystal [2]. Systems of hard spheres near their jamming point exhibit hyperuniformity, suggesting that the suppression of density fluctuations is a signature of mechanical criticality. At the largest accessible scales, the distribution of galaxies in cosmological surveys appears nearly hyperuniform, hinting at long-range correlations established in the early universe. Quasicrystals — aperiodic materials with long-range orientational order but no translational periodicity, first observed in an aluminium-manganese alloy by Shechtman and coworkers [3] — are naturally hyperuniform because their long-range aperiodic structure prevents macroscopic density imbalances. From an engineering perspective, computationally designed disordered hyperuniform networks have been shown to support photonic band gaps rivaling those of periodic photonic crystals, offering a route to isotropic wave-guiding that does not require strict periodicity [2].

The formal definition of hyperuniformity, due to Torquato and Stillinger [1], is that the structure factor $S(k)$ — the Fourier-space measure of density fluctuations at spatial wavelength $2\pi/k$ — satisfies $S(k) \to 0$ as $k \to 0$. The *rate* of this vanishing defines a spectrum of hyperuniformity. For systems where $S(k) \sim |k|^\alpha$ near $k = 0$, the exponent $\alpha > 0$ quantifies the degree of suppression: larger α means faster vanishing and more strongly reduced fluctuations at long wavelengths. Three universality classes arise from the asymptotic behavior of $\sigma^2(R)$ [2]. Class I ($\alpha > 1$) is the strongest: the number variance is bounded at large R, approaching a constant plateau. Perfect crystals and many quasicrystals belong to this class, as do stealthy hyperuniform systems for which $S(k) = 0$ over an entire exclusion zone near $k = 0$. Class II ($\alpha = 1$) is the marginal case where $\sigma^2(R)$ grows logarithmically; the period-doubling substitution chain is a one-dimensional example. Class III ($0 < \alpha < 1$) gives power-law variance growth strictly slower than the Poisson baseline; certain substitution chains with non-Pisot eigenvalues fall here.

One spatial dimension is a natural test bed for hyperuniformity theory. Point patterns in 1D can be generated deterministically to sizes $N \sim 10^7$ by iterating substitution rules, exact analytical benchmarks exist, and the connection between a pattern's algebraic structure and its hyperuniformity class is particularly transparent. Yet the 1D landscape of known Class I patterns is sparse. The survey in Ref. [2] lists only a handful of examples, and the interval $1 < \alpha < 2$ was previously unoccupied by any known 1D quasicrystal. As we show below, the reason is algebraic: all 2×2 metallic-mean substitution matrices are unimodular ($|\det M| = \pm 1$), which forces $\alpha = 3$ exactly [4]. Filling the $1 < \alpha < 2$ gap requires moving outside this family.

In this work, we characterize 27 one-dimensional point patterns spanning all three hyperuniformity classes. The patterns include the five metallic-mean quasicrystals (Fibonacci, Silver, Bronze, Copper, and Nickel chains), all of which have $\alpha = 3$ by the determinant constraint; uncorrelated

random lattice (URL) perturbations, which realize $\alpha = 2$ exactly and serve as key validation benchmarks; stealthy hyperuniform configurations generated by collective coordinate optimization; and several substitution chains that realize Class II and Class III behavior. The central new result is the identification of the Bombieri-Taylor cubic substitution [5] as a Class I quasicrystal with $\alpha = 1.545$, the first known example with $1 < \alpha < 2$ [4]. We also report surface-area coefficients $\bar{\Lambda}$ for the Silver, Bronze, Copper ($\bar{\Lambda} = 0.293$), and Nickel ($\bar{\Lambda} = 0.310$) chains; the Copper and Nickel values are reported here for the first time.

The remainder of this paper is organized as follows. Section 2 develops the theoretical framework: the number variance and its relation to the pair correlation function, the structure factor and hyperuniformity exponent, the surface-area coefficient $\bar{\Lambda}$, and the eigenvalue formula for substitution tilings. Section 3 describes the numerical methods and validation benchmarks used throughout this work. Results and conclusions will follow in subsequent sections.

2 Theoretical Background

2.1 Number Variance

To quantify how density fluctuations depend on scale, we examine how many points fall inside a window of half-width R placed at a random position along a one-dimensional point pattern of mean linear density ρ. The variance of this count across window placements is the *number variance* $\sigma^2(R)$. For a Poisson (completely uncorrelated) point process with density ρ, the variance grows linearly with window size: $\sigma^2_{\text{Poisson}}(R) = 2\rho R$ [1]. This Poisson scaling is the baseline against which all other patterns are measured — it reflects the fact that in an uncorrelated system, each region of space is independent, so larger windows simply accumulate more independent uncertainty.

For a general stationary ergodic point process, $\sigma^2(R)$ can be expressed in terms of the pair correlation function $g_2(r)$, which encodes the probability of finding a second point at distance r from a reference point relative to the uncorrelated expectation [1]:

$$\sigma^2(R) = 2\rho R \left[1 + 2\rho \int_0^{2R} \left(1 - \frac{r}{2R} \right) \left[g_2(r) - 1 \right] dr \right]. \tag{1}$$

The first term, $2\rho R$, is the uncorrelated (Poisson) contribution. The integral represents the correction from inter-point correlations: where $g_2(r) > 1$ (clustering), the variance increases; where $g_2(r) < 1$ (anticorrelation), the variance decreases. A pattern is **hyperuniform** if the integral correction exactly cancels the Poisson term in the large-R limit [1]:

$$\lim_{R \to \infty} \frac{\sigma^2(R)}{2\rho R} = 0. \tag{2}$$

Physically, this requires that points be anticorrelated at large separations in precisely the right amount: $\int_0^\infty [g_2(r) - 1]\, dr = -1/(2\rho)$.

Based on how $\sigma^2(R)$ grows asymptotically, hyperuniform patterns fall into three universality classes [2]:

- **Class I** ($\alpha > 1$): $\sigma^2(R)$ approaches a finite (though possibly oscillating) constant at large R. These patterns exhibit the strongest suppression of density fluctuations. Perfect crystals and quasicrystals generated by Pisot substitution rules belong here.

- **Class II** ($\alpha = 1$): $\sigma^2(R) \sim C \ln R$. The variance grows without bound, but only logarithmically. The period-doubling chain is a 1D example.

- **Class III** ($0 < \alpha < 1$): $\sigma^2(R) \sim R^{1-\alpha}$, a power-law growth strictly slower than the Poisson baseline. Aperiodic chains with non-Pisot eigenvalue spectra can realize this behavior.

Non-hyperuniform patterns ($\alpha \leq 0$) have $\sigma^2(R) \sim R$ or faster.

2.2 Structure Factor and Hyperuniformity Exponent

The same information encoded in $\sigma^2(R)$ is also captured by the *structure factor* in Fourier space. For a point pattern with N points at positions $\{x_j\}$,

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{ikx_j} \right|^2 . \tag{3}$$

Physically, $S(k)$ measures the intensity of density fluctuations at spatial wavelength $2\pi/k$. For a Poisson process $S(k) = 1$ at all k; for a perfect crystal $S(k) = 0$ for all $|k|$ below the first reciprocal lattice vector. The two observables are related by an exact integral [1]:

$$\sigma^2(R) = \frac{\rho}{\pi} \int_0^\infty S(k) \left(\frac{2\sin(kR)}{k} \right)^2 dk. \tag{4}$$

Hyperuniformity requires $S(k) \to 0$ as $k \to 0$, which causes the integrand to vanish faster than for Poisson and yields sub-linear variance growth at large R.

For systems where $S(k)$ approaches zero as a power law,

$$S(k) \sim B|k|^\alpha \quad \text{as } k \to 0, \tag{5}$$

the exponent $\alpha > 0$ is the *hyperuniformity exponent*. Larger α implies more strongly suppressed fluctuations at long wavelengths. For quasicrystals, the structure factor consists of dense Bragg peaks rather than a smooth continuous envelope, so the exponent is extracted from the integrated intensity $Z(k) = 2\int_0^k S(q)\,dq \sim k^{1+\alpha}$ near $k = 0$ [4]. In practice, α is also estimated from the asymptotic slope of $\sigma^2(R)$ on a log-log scale, providing an independent numerical check.

2.3 Surface-Area Coefficient $\bar{\Lambda}$

Within Class I, all patterns have bounded number variance, but two systems with the same α can differ substantially in their residual local disorder. The *surface-area coefficient* $\bar{\Lambda}$ is the secondary metric that resolves this ambiguity and provides an absolute ranking within Class I.

For Class I systems in one dimension, $\sigma^2(R)$ does not simply converge to a constant but typically oscillates about a plateau, particularly for quasicrystals with dense Bragg spectra. A pointwise limit is therefore not well-defined in general. Instead, $\bar{\Lambda}$ is defined as the long-range running average [2, 1]:

$$\bar{\Lambda} = \lim_{R\to\infty} \frac{1}{R} \int_0^R \sigma^2(R')\,dR'. \tag{6}$$

This running average converges to the mean level of the oscillating variance plateau, and reduces to the ordinary limit $\lim_{R\to\infty} \sigma^2(R)$ whenever that limit exists. For systems where the variance oscillates without converging, Eq. (6) extracts a unique, well-defined amplitude.

The name "surface-area coefficient" refers to the d-dimensional interpretation: for a Class I pattern in d dimensions, the variance scales at large R as $\sigma^2(R) \sim 2\bar{\Lambda}\,R^{d-1}$, where R^{d-1} is proportional to the surface area of the observation window. In $d = 1$ this reduces to $\sigma^2(R) \to \bar{\Lambda}$ (a

constant), consistent with the definition Eq. (6). The quantity $\bar{\Lambda}$ is the primary ranking metric within Class I: the perfect integer lattice achieves the global minimum $\bar{\Lambda} = 1/6$, proved exact in Ref. [1], and any deviation from perfect periodicity strictly increases $\bar{\Lambda}$. For the Fibonacci chain, Ref. [6] established $\bar{\Lambda} \approx 0.201$ by an analytical theta-function method.

2.4 Substitution Tilings and the Eigenvalue Formula

A one-dimensional *substitution tiling* is an aperiodic sequence of tiles generated by repeatedly applying a fixed rewriting rule to each letter of a finite alphabet. Starting from a seed word, each letter is replaced by a fixed string of letters; iteration produces an increasingly long sequence that, when each tile is assigned a physical length and the tiles are stacked end to end, defines a one-dimensional point pattern. These tilings provide some of the cleanest examples of quasicrystalline order: they are fully deterministic, can be generated to arbitrary length by iteration, and their algebraic properties encode their hyperuniformity class in a precise and computable way.

The growth of a substitution tiling is governed by its *substitution matrix* M, whose entry M_{ij} counts how many tiles of type j appear in the substituted image of a single tile of type i. For the Fibonacci chain with alphabet $\{S, L\}$ and rules $S \to L$, $L \to LS$, the substitution matrix is

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \tag{7}$$

with eigenvalues $\lambda_1 = \tau = (1 + \sqrt{5})/2 \approx 1.618$ (the golden ratio) and $\lambda_2 = -1/\tau \approx -0.618$. The Perron-Frobenius theorem guarantees that the largest eigenvalue λ_1 is positive and real, and its eigenvector gives the asymptotic relative frequencies of each tile type. Tile lengths are assigned by requiring that inflating every tile by λ_1 and then applying the substitution leaves the total domain length invariant, which fixes the ratio $L/S = \lambda_1$. For the Fibonacci chain this gives $L = \tau S$: the long tile is τ times the short tile.

The key result connecting the matrix spectrum to hyperuniformity is due to Oğuz et al. [7, 4]: for a substitution tiling with Perron-Frobenius eigenvalue λ_1 and second-largest-magnitude eigenvalue λ_2,

$$\alpha = 1 - \frac{2 \ln |\lambda_2|}{\ln |\lambda_1|}. \tag{8}$$

This formula is the central predictive tool of this work: α is determined entirely by the substitution matrix spectrum, with no simulation required. The tiling is hyperuniform when $\alpha > 0$, which requires $|\lambda_2| < |\lambda_1|^{1/2}$. The physical interpretation is clear: $|\lambda_1|$ sets the inflation rate under substitution, while $|\lambda_2|$ controls how quickly spatial correlations across tile boundaries decay. A larger ratio $|\lambda_2|/|\lambda_1|^{1/2}$ means slower decorrelation and weaker hyperuniformity.

For the metallic-mean chains and for the Bombieri-Taylor cubic substitution, the largest eigenvalue λ_1 is a *Pisot-Vijayaraghavan (PV) number*: an algebraic integer greater than one all of whose algebraic conjugates lie strictly inside the unit circle ($|\lambda_j| < 1$ for $j \geq 2$) [5, 4]. The PV property guarantees a pure Bragg diffraction spectrum — the tiling is a true quasicrystal in the crystallographic sense, with delta-function peaks in $S(k)$ at a dense set of wavevectors. Equation (8) also reveals the structural reason why all 2×2 metallic-mean matrices are locked to $\alpha = 3$ and why accessing intermediate exponents requires a higher-rank substitution matrix.

3 Methods

We characterize each point pattern using three complementary observables: the number variance $\sigma^2(R)$, the structure factor $S(k)$, and the diffusion spreadability $\mathcal{S}(t)$ [8, 9]. These are computed

from finite point patterns generated to sizes large enough that finite-size effects are negligible on the window and wavenumber scales of interest. This section describes how each pattern type is generated, how each observable is computed, and how the results are validated against exact benchmarks.

3.1 Pattern Generation

Substitution tilings. Each substitution tiling is generated by iterating the substitution rule on a seed word, continuing until the total point count exceeds a target N_{target}. At each iteration every tile of type i is replaced by the string of tiles prescribed by row i of the substitution matrix M; after n iterations the total tile count grows approximately as λ_1^n. Tile lengths are set by the Perron-Frobenius eigenvector of M: if $v_i > 0$ is the i-th component, the length of tile type i is $\ell_i = v_i / \min_j v_j$, so that the shortest tile always has unit length. Points are placed at tile boundaries. The resulting pattern has N points on a domain of length $L \approx N/\rho$, where the mean density is $\rho = 1/\bar{\ell}$ and $\bar{\ell} = \sum_i f_i \ell_i$ is the frequency-weighted mean tile length computed from the asymptotic tile frequencies f_i given by the normalized Perron-Frobenius eigenvector. For the metallic-mean chains, N ranges from 1.5×10^4 to $\sim 4 \times 10^7$ depending on convergence requirements; for the Bombieri-Taylor cubic, 25 substitution iterations yield $N \approx 9.5 \times 10^6$.

Uncorrelated random lattices. The uncorrelated random lattice (URL) is generated by taking the integer lattice $\{1, 2, \ldots, N\}$ and displacing each site independently by a random variate drawn from a uniform distribution parameterized by amplitude a. Realizations at $a = 0.1$, 0.5, and 1.0 are used; for each, the variance is computed from a single large realization ($N = 10^6$) and compared with the exact formula $\bar{\Lambda} = (1 + a^2)/6$ derived by Klatt, Kim, and Torquato [10].

Stealthy configurations. Stealthy hyperuniform configurations, for which $S(k) = 0$ for all $|k| < K^*$, were generated by a collaborating group using collective coordinate optimization [11]. The ensemble consists of approximately 4,316 independent configurations per value of the stealthiness ratio $\chi \in \{0.1, 0.2, 0.3\}$, each containing $N = 2000$ points at mean density $\rho = 1$. Rather than a single large system, the high-statistics ensemble average over thousands of configurations suppresses sample-to-sample noise and provides reliable estimates of both $S(k)$ and $\bar{\Lambda}$.

3.2 Number Variance and Surface-Area Coefficient

For a pattern of N points on a domain of length L, the number variance $\sigma^2(R)$ at window half-width R is estimated by Monte Carlo sampling. We place $N_{\text{samp}} = 30{,}000$ windows of half-width R at positions drawn uniformly at random along the domain, count the number of points n_i in each window, and compute $\sigma^2(R) = \langle n^2 \rangle - \langle n \rangle^2$. The window half-width is swept over a logarithmic grid from $R_{\min} = 1/\rho$ (one inter-point spacing) to $R_{\max} = L/4$; the upper bound ensures that every window placement is fully contained within the domain and that the sampling is not dominated by boundary effects. At each R, the estimator is essentially unbiased because the window placements are independent of the point pattern itself.

The surface-area coefficient $\bar{\Lambda}$ is extracted as the running average of the variance curve [1],

$$\bar{\Lambda}(R_{\max}) = \frac{1}{R_{\max}} \int_0^{R_{\max}} \sigma^2(R')\, dR'. \tag{9}$$

For Class I patterns, this running average converges to a plateau as $R_{\max}$ increases, providing the estimate of $\bar{\Lambda}$. Convergence is monitored by checking that Eq. (9) varies by less than 1% as $R_{\max}$ is doubled; for all substitution tilings and URL patterns studied here, this criterion is met well within the available domain length. For stealthy configurations, $\bar{\Lambda}$ is computed from the ensemble-averaged

variance curve at $R_{\max} = N/(4\rho)$, where $N = 2000$ provides sufficient range for convergence at all three χ values studied.

3.3 Structure Factor and Hyperuniformity Exponent

The structure factor $S(k)$ is evaluated on the discrete wavenumber grid $k_n = 2\pi n/L$ for integers $n = 1, 2, \ldots, N/2$ using the direct sum in Eq. (3), computed via a fast Fourier transform. The low-k region, which determines the hyperuniformity class and exponent, is sampled at fine wavenumber resolution and the $S(k_n)$ values are logarithmically binned to reduce scatter before fitting.

For patterns with a smooth $S(k)$ near $k = 0$ (URL, stable distributions, stealthy), the exponent α is extracted from an unweighted linear fit to $\log S(k)$ versus $\log k$ over the wavenumber range where a clear power law is evident. For substitution tilings, $S(k)$ consists entirely of isolated Bragg peaks rather than a smooth envelope; a direct log-log fit to peak heights is unreliable. Instead, we follow Oğuz et al. [4] and fit the cumulative intensity

$$Z(k) = 2\int_0^k S(q)\,dq \sim C\,k^{1+\alpha}, \tag{10}$$

which is a smooth, monotonically increasing function of k even when $S(k)$ is a discrete set of peaks. An unweighted linear fit to $\log Z(k)$ versus $\log k$ over a range spanning at least one decade in k yields α with uncertainty estimated from the fit residuals. As an independent cross-check, α is also estimated from the log-log slope of $\sigma^2(R)$ at large R; for all Class I patterns studied, the two methods agree to within 5%.

3.4 Diffusion Spreadability

The diffusion spreadability $\mathcal{S}(t)$ quantifies the fraction of space not yet reached by a species diffusing from the point pattern at time t [8, 9]. In one dimension and for a two-phase medium with packing fraction ϕ_2, the spreadability is related to $S(k)$ via

$$\mathcal{S}(t) = \phi_2^2 + \frac{1}{\pi}\int_0^\infty S(k)\,e^{-2k^2 t}\,dk. \tag{11}$$

For Class I and Class II hyperuniform patterns, the excess spreadability $E(t) = \mathcal{S}(t) - \phi_2^2$ decays as a power law at long times: $E(t) \sim t^{-(1+\alpha)/2}$ [8], while stealthy configurations exhibit exponential decay. We evaluate $\mathcal{S}(t)$ numerically by discretizing the integral over the precomputed structure factor values $\{S(k_n)\}$ at a two-phase packing fraction $\phi_2 = 0.35$ [12]. In practice, we use spreadability primarily as an independent qualitative consistency check on the class assignment. For substitution tilings with dense Bragg spectra — in particular the Bombieri-Taylor cubic, which has 32,517 Bragg peaks in $k < 50$ — the integral (11) requires extremely long times to converge, and the eigenvalue formula Eq. (8) is used instead for precise α estimates.

3.5 Validation Benchmarks

Before applying these methods to the patterns under study, all three observables are validated against exact analytical benchmarks.

Poisson process. For a Poisson point process at density $\rho = 1$ and system size $N = 10^6$, the exact results are $\sigma^2(R) = 2\rho R$ and $S(k) = 1$ for all k. Numerical estimates reproduce the Poisson variance to within 1.1% across three decades of window size and $S(k) = 1.000 \pm 0.002$ at all sampled

wavenumbers, confirming that the window-sampling estimator and the FFT-based structure factor introduce no systematic bias.

Uncorrelated random lattice. For the URL at amplitudes $a = 0.1$, 0.5, and 1.0, the exact surface-area coefficient is $\bar{\Lambda} = (1 + a^2)/6$, derived analytically in Ref. [10]. Numerical estimates agree with this formula to within 0.3% at all three amplitudes, validating both the number-variance estimator and the running-average extraction of $\bar{\Lambda}$.

Fibonacci chain. For the Fibonacci quasicrystal, Zachary and Torquato [6] computed $\bar{\Lambda} = 0.20110$ analytically using a theta-function representation of the pair correlation function. Our Monte Carlo estimate at $N \approx 1.5 \times 10^7$ gives $\bar{\Lambda} = 0.200 \pm 0.001$, consistent with the analytical value at the level of numerical precision; this provides a cross-check on the method for quasiperiodic patterns where no simple closed-form formula exists.

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